NumPy: The Mathematical Foundation

NumPy provides the mathematical foundation for scientific computing in Python. It offers powerful array operations and mathematical functions that execute much faster than native Python code.

Real-world application: Renaissance Technologies, the legendary quantitative hedge fund with one of the best performance records in history, uses numerical libraries like NumPy for their tensor calculations in statistical arbitrage models.

Installation:

```
pip install numpy
```

Simple example:

```python
import numpy as np

# Create an array of stock prices
prices = np.array([100, 102, 104, 103, 105])

# Calculate daily returns
returns = np.diff(prices) / prices[:-1] * 100

print(f"Daily returns: {returns}%")

```

pandas: Your Data Manipulation Swiss Army Knife

pandas is perhaps the most important library for financial data analysis. It provides DataFrame objects (similar to spreadsheets or SQL tables) that make working with financial time series data intuitive and efficient.

Real-world application: BlackRock, the world's largest asset manager with over $9 trillion under management, uses pandas extensively in their Aladdin platform for risk analysis and portfolio management.